

# The Cost Geometry of Belief

*finite-resource inference under noisy observation*

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*preliminary version — comments welcome*

**Abstract.** We equip the space of beliefs with a *cost geometry* (what it costs to *pass* from one belief to another): optimal transport in Wasserstein space, *reweighted* conformally by Fisher information (the price of the precision at stake), distinct from the Fisher–Rao metric. In the setting we consider, a finite machine’s digital twin of a system observes the territory through finite, noisy sensors, and we model its coherent output as a *belief*: a probability density over states, the Bayes posterior. Certainty — the *perfect twin* — is denied twice, by observation and by physics, both read off the Fisher information. On the conformal class, essentially location-scale, three results emerge, all *invariants of one change of cost unit*. A *wall*: a well-posed inference rejects certainty to infinite distance as soon as the cost dominates the Fisher information (necessity conjectured beyond power laws). An *honesty*: an honest (eikonal) cost — each nat the same length everywhere — selects the geometries proportional to the Fisher information. A *rigidity*: these geometries are hyperbolic, and the Stam bound crowns the Gaussian, the *most hyperbolic* location-scale belief. Changing the unit dilates the geometry yet preserves the wall, the curvature ordering, and the Gaussian’s extremality: an absolute cost says nothing, only relative cost carries meaning, the value  $-1/4$  being one of its images. The cost of reaching a given *precision* then has a geometric floor diverging at certainty. Thermodynamics fixes the cost unit and motivates this framework; the results are geometric, in nats.

## 1 Introduction

A *digital twin* (Grieves and Vickers, 2017) maintains, inside a machine, an image of a real system — a territory, a city, a structure — in order to track or anticipate it. It is often conceived as a *replica*: a deterministic copy that data refresh. In the setting we consider, however, the machine that realizes it is *finite* and accesses the system only through finite, noisy channels — whether a physical sensor or a *finite agent* (a human included), each itself noisy. Its coherent output is then a *belief*: a probability density over the possible states, produced by Bayesian inference, the coherent framework for reasoning under uncertainty (Cox, 1946; Jaynes, 2003; MacKay, 2003). Practice already does this — ensembles, data assimilation, Kalman filters (Kalman, 1960) — often without naming it.

To such a belief, *certainty* — the perfect twin, without uncertainty — is denied *twice*. First by observational finiteness: perfect precision would require unbounded information, which the Cramér–Rao inequality rules out (Cover and Thomas, 2006, Th. 11.10.1). Then by physical finiteness: erasing information dissipates energy, by the Landauer principle (Landauer, 1961). These two refusals are read off from one and the same quantity, the Fisher information (Fisher, 1925). Finiteness does not invent the belief — the belief is born of inference —; it only keeps it from freezing into certainty. Certainty is therefore a boundary that no finite system crosses.

The belief thus laid down, what remains to describe is *geometric*: the shape of the space where these beliefs live and move — its distances, its boundary (certainty) and its curvature. The cost

measures a *change* of belief, the passage from one to the other: moving probability mass (optimal transport, at constant mass), whose length is *reweighted*, conformally, by the Fisher information (definition 2.1); the more *precise* the belief is there, the more dearly each step is paid. This information measures here the *square of the slope* of the entropy. The cost is thus measured up to a unit — only relative gaps have a meaning (theorem 4.21). The geometric toolbox nevertheless already exists: information geometry endows the laws with the Fisher–Rao metric (Chentsov, 1972; Amari, 2016), and optimal transport metrizes their displacement (Otto, 2001); geometric thermodynamics (Sivak and Crooks, 2012; Aurell et al., 2011; Ito, 2023) and Bayesian mechanics (Friston, 2019; Sakthivadivel, 2022) already tie dissipation (the former) and an agent (the latter) to a geometry of belief. But these formalisms measure *distinguishability*, or dissipation *on a given geometry*: none, to our knowledge, makes finiteness the constraint that *re-metrizes* the space of beliefs into a geometry of *cost*, endowed with an uncrossable boundary and a curvature.

We *characterize* these cost geometries — on the conformal class, and essentially on location-scale families — by three results. A *wall*: a *well-posed* inference — in the sense of existence and stability, uniqueness remaining open — requires certainty to be rejected to infinite distance. The domination of the cost over the Fisher information ensures it: sufficient in every dimension, necessary at the boundary on power laws, the general case conjectured. An *honesty*: requiring that each unit of knowledge cost everywhere the same length selects the geometries proportional to the Fisher information. A *rigidity*: on location-scale families, these geometries are hyperbolic, their curvature there reading the Fisher information *of the base*, and the Stam bound (Stam, 1959) makes the Gaussian there the most hyperbolic belief. This yields, by the wall, a cost floor for reaching a given precision: intrinsic to the geometry, independent of the inference procedure, it diverges as certainty is approached. Thermodynamics *motivates* this framework, through the entropy–energy reading of Landauer; the bridge to the geometry is the Otto identity — the square of the slope of the entropy there equals the Fisher information —, a transport fact. No proof depends on it: all are geometric, in nats.

A finite twin measures its cost in a unit; changing it dilates the distances, divides the curvature by the same factor, and preserves the wall, the ordering of curvatures and the extremality of the Gaussian. This is the *gauge invariance of cost*, or *relativity of cost* (theorem 4.21). Thermodynamics fixes this unit — at the wall, one nat costs  $k_B T$ .

This frontier has a direct algorithmic reading: a Kalman filter, which maintains its uncertainty, revises itself at finite cost, whereas a network frozen on a certainty lives at the boundary and must be retrained. These two cases are *exact* reductions of the framework (box 1).

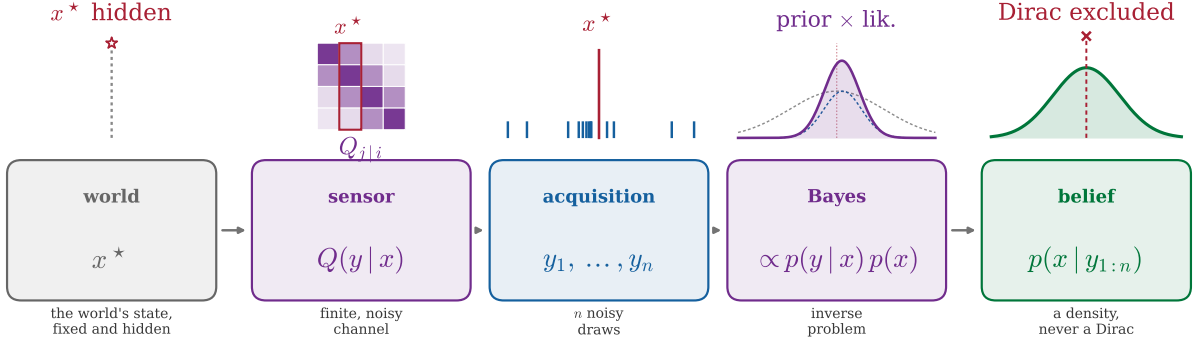
## 2 Setting and problem

We first state the question, then build the formal framework that makes it precise.

**Problem 1** (the central problem). *Characterize the geometry of the space of beliefs that a finite agent, inferring a fixed and noisy world through finite sensors, can inhabit: its distances (the cost of changing one’s mind), its boundary (certainty) and its curvature.*

Let us formalize. An agent has no access to the state of the world  $x$ , assumed fixed: a sensor interposes itself, and the agent receives only observations  $y$ . Bayesian inference answers this, through Bayes’ theorem (MacKay, 2003, p. 25, §2.3). We adopt a static reading: the state  $x$  is fixed, the agent has a finite number of observations, and we ask what it may, at a given instant, legitimately believe.

We model the sensor as a finite-resolution and noisy channel, whose transition matrix  $Q_{j|i} = P(y = b_j \mid x = a_i)$  gives the probability of observing  $b_j$  when the world is in state  $a_i$  (MacKay, 2003, p. 147, eq. 9.1): it transmits a version of the state filtered by  $Q$ . The output of inference is then a *belief*, the posterior  $p(x \mid y) \propto p(y \mid x)p(x)$  (MacKay, 2003, p. 148, eq. 9.4); inferring,



**Figure 1: The inference chain: from a noisy world to a belief** (figure computed numerically). The state  $x^*$  (fixed, hidden); the finite and noisy sensor  $Q(y|x)$  ( $Q_{j|i} = P(y=b_j | x=a_i)$ ); the  $n$  draws  $y_1, \dots, y_n$ , scattered around  $x^*$  by the noise; the belief  $p(x|y_{1:n}) \propto p(y|x)p(x)$  obtained by Bayes. The output is a *density* (of entropy  $H = -\int p \ln p$ ); certainty, the barred centered Dirac, is excluded from it.

decoding a channel and perceiving are one and the same inverse problem (MacKay, 2003, p. 156, §9.7). The belief is thus a *density* over the states of the world.

To this belief certainty is denied *twice*, and both denials speak the same language, that of Fisher information  $J$ . First through observational finiteness. Specifying the world to a resolution  $\delta$  costs  $h(X) + \ln(1/\delta)$  nats, a quantity that diverges as  $\delta \rightarrow 0$  (MacKay, 2003, p. 180); the same barrier reads in estimation: by Cramér–Rao  $\text{Var}(\hat{\theta}) \geq 1/J(\theta)$ , a zero variance would require  $J = \infty$  (Cover and Thomas, 2006, Th. 11.10.1); and since the channel capacity is finite, processing creates no information,  $I(x; \hat{x}) \leq I(x; y) \leq NC$  (Cover and Thomas, 2006, Th. 2.8.1). Then through physical finiteness: erasing information dissipates at least  $k_B T \ln 2$  per bit (Landauer, 1961; Bennett, 1982), verified experimentally (Bérut et al., 2012), and the information capacity of a bounded region is itself bounded (Bekenstein, 1981). Reaching certainty would require infinite energy and memory. The density comes from inference; the two forms of finiteness only exclude its contraction to a Dirac. figure 1 summarizes this chain, from the noisy world to the belief-density.

We formalize the framework in three choices — the object (the belief-density), the arena (the Wasserstein transport) and the cost (the conformal relief) —, laid out below, each with what it discards (the point, the Fisher–Rao tensor, the non-conformal deformation, the non-uniform price; table 1).

**Definition 2.1** (arena, belief, entropy, Fisher). The *arena* is  $(\mathcal{P}_2(\mathbb{R}^n), \mathcal{W}_2)$ , the Wasserstein space of probability measures with finite second moment. A *belief* is a density  $p$ ; its entropy (Shannon, nats) is  $H(p) = -\int p \ln p dx$ , its knowledge  $-H(p)$ , its Fisher information  $J(p) = \int |\nabla p|^2/p \in [0, +\infty]$ . We write  $\text{Ent} := -H$  and  $\partial_\infty := \{H = -\infty\}$  the *boundary of certainties* (singular measures, including all Diracs). Gaussian dictionary ( $n = 1$ ):  $H = \frac{1}{2} \ln(2\pi e \sigma^2)$ ,  $J = 1/\sigma^2$ .

**Definition 2.2** (regularity class (*setting*)).  $\mathcal{R} \subset \mathcal{P}_2(\mathbb{R}^n)$ :  $C^1$  densities, strictly positive, with finite  $m_2$  and  $J$ . It is the interior domain;  $\partial_\infty$  is excluded from it.

**Assumption 1** (A1 — object (*setting*)). The belief is a density of class  $\mathcal{R}$  (definition 2.2), a point of the arena. Finiteness *closes off the Dirac* ( $\partial_\infty$  excluded: infinite precision would make  $J$  infinite there); the continuous  $C^1$  class is a working idealization.

*Remark 2.3* (two operational constraints (*setting*)). Two constraints drive the characterization. (a) *Boundary excluded*:  $\partial_\infty$  (infinite Fisher information, definition 2.1) is out of budget, the belief lives in the interior ( $\mathcal{R}$ ). (b) *Well-posed*: inference — minimization over the arena — must admit an *interior* minimizer, with no collapse toward  $\partial_\infty$ . (The boundary is costly because  $J$  diverges there;  $J$  is here the slope of  $-H$ , not the Fisher–Rao tensor — remark 2.5.)

**Definition 2.4** (plausibility and relief (*setting*)). The plausibility of a belief — no longer the posterior  $p(x | y)$  over the states of the world, but a degree over the beliefs themselves — is modeled by a Gibbs measure on the space of beliefs (Jaynes, 2003),

$$\Pi \propto \mu e^{-U}, \quad U = -\ln \frac{d\Pi}{d\mu} \geq 0,$$

$\mu$  being a reference measure. The *relief*  $U$  is the negated log-density: an *improbability* (least commitment). The boundary of certainties ( $-H = +\infty$ ) is all the less plausible the more the relief grows there; no form of  $U$  is fixed here.

$\Pi$  is a measure on the space of beliefs: a density, not an evolution. The relief  $U$  remains *generic* — lower semi-continuous,  $\geq 0$  —, never a fixed functional form.

**Assumption 2** (A2 — measure (*setting*)). Knowledge is measured by the Shannon entropy  $-H$  (Shannon, 1948). (*It is the entropy that Landauer prices, and the one for which the Otto identity assumption 4 equals  $\sqrt{J}$ : the identity relating the slope of  $-H$  to Fisher information is specific to the Shannon entropy; it does not extend as is to a Rényi one.*)

What is costly to the agent is to *change* belief: to move probability mass. We model this cost as the Wasserstein optimal transport (assumption 3).

**Assumption 3** (A3 — arena (*setting, canonical*)). The geometry deforms *transport*: arena  $(\mathcal{P}_2, \mathcal{W}_2)$ . (*Canonical choice, not unique; cf. remarks 2.5 and 7.1.*)

*Remark 2.5* (the dual role of  $J$  — slope of  $-H$ , not tensor).  $J$  occupies two opposite places: (a) *metric tensor* (Fisher–Rao, Čencov) — *not us*; (b) *squared slope of a function*,  $J = \|\nabla_{\mathcal{W}_2} H\|^2$  (assumption 4) — *us*. We take  $J$  as the slope of  $-H$  on the transport arena.

We take the relief  $U$  to warp the transport metric *conformally*:  $\tilde{g} = 2(e + U) g_{\mathcal{W}_2}$ . The candidate class  $\mathcal{U}$  gathers all admissible reliefs.

**Definition 2.6** (space of candidates (*setting*)). A *relief* is a function  $U : \mathcal{P}_2(\mathbb{R}^n) \rightarrow [0, +\infty]$ . The *space of candidates* is

$$\mathcal{U} := \{ U : \mathcal{P}_2(\mathbb{R}^n) \rightarrow [0, +\infty] \mid U \text{ is } \tau\text{-sequentially l.s.c.} \},$$

$\tau$  being the narrow convergence. To each  $U \in \mathcal{U}$  and each level  $e \geq 0$  is associated the *cost geometry* (2: unit;  $e > 0$  for licit reparametrization and separation of points,  $e = 0$  for the eikonal/limiting case, conformal metric on  $\{U > 0\}$ )

$$\tilde{g}_{e,U} := 2(e + U) g_{\mathcal{W}_2}, \quad \ell_e(\gamma) := \int_0^T \sqrt{2(e + U(\gamma_t))} |\dot{\gamma}|(t) dt, \quad d_e := \inf \ell_e. \quad (2.1)$$

The candidate class is *conformal*: we deform  $g_{\mathcal{W}_2}$  by a scalar. Non-conformal deformations are out of class (remark 6.1).

The characterization will bear on  $U \in \mathcal{U}$ , with three nested subclasses: the *wall class*  $\mathcal{W}$  (definition 4.9), the *well-posed class*  $\mathcal{C}$  (definition 4.8) and the *Fisher family*  $\mathcal{F}$  (definition 4.13),  $\mathcal{F} \subset \mathcal{C} \subset \mathcal{W} \subset \mathcal{U}$ .

**The framework, in plain terms.** The problem we attack boils down to five commitments, inherited from the introduction.

- **A finite agent.** Bounded energy, memory and degrees of freedom; for it, information is physical (Landauer, Bekenstein).
- **A fixed world, seen through a noisy sensor.** The state  $x$  is unknown and fixed; the agent accesses it only through a noisy channel  $Q(y | x)$  of finite resolution.

**Table 1: The framework in one table** — the six commitments and the alternative that each rejects.

| Element                       | Choice (& why)                                                                   | Excludes                        |
|-------------------------------|----------------------------------------------------------------------------------|---------------------------------|
| Object (assumption 1)         | density $\in \mathcal{R}$ (Dirac = infinite precision $\Rightarrow J = \infty$ ) | the point / Dirac               |
| Plausibility (definition 2.4) | $\Pi \propto \mu e^{-U}$ , relief $U$ = improbability                            | fixed form $U$                  |
| Arena (assumption 3)          | Wasserstein $\mathcal{W}_2$ (we bill the <i>change</i> )                         | Fisher–Rao (distinguishability) |
| Measure (assumption 2)        | $-H$ Shannon (Landauer; pivot $\sqrt{J}$ )                                       | Rényi & others                  |
| Cost (definition 2.6)         | conformal relief $2(e + U)g_{\mathcal{W}_2}$                                     | non-conformal (C1)              |
| Eikonicity (definition 4.12)  | constant price per nat                                                           | $\alpha \neq 1$                 |

- **A belief is a density.** The output of inference is the posterior  $p(x | y)$ , never a point.
- **Certainty is out of reach — twice.** The Dirac (zero variance) is excluded by two forms of finiteness both read in Fisher information  $J$ : *observational* ( $J \rightarrow \infty$ , finite capacity) and *physical* (infinite energy).
- **Cost reads geometrically.** What is costly is to *change* belief — a Wasserstein transport — and we *choose* to deform it conformally by a cost relief  $U$ . The precise form of this candidate class ( $\tilde{g} = 2(e + U)g_{\mathcal{W}_2}$ , where the “2” is a unit and  $e$  a level) is laid out in Section 2.

From here on, the question is geometric in nature.

### 3 Overview: results, intuition, and classes

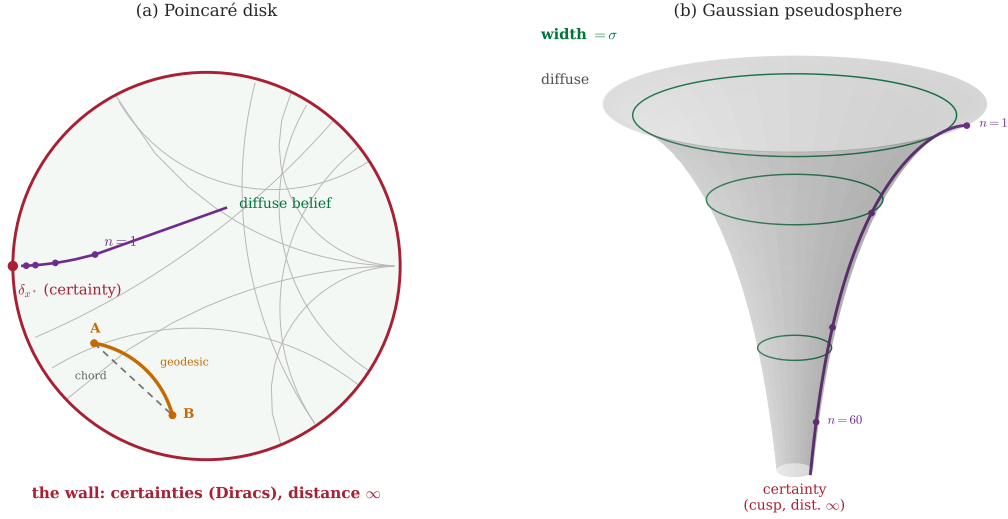
This section first states the three results, illuminates them geometrically (a table then a figure), and maps their classes; the proofs come in Section 4.

**Contributions.** We characterize this space by three results, proved in metric geometry; figure 3 maps their nested classes.

- **The wall** (theorem 4.10). A well-posed inference requires that certainty be pushed to infinite distance; this wall holds *if* the relief dominates the Fisher information (unconditional sufficiency, in every dimension), domination *necessary at the boundary* on the power family  $\{cJ^\alpha\}$  ( $\Leftrightarrow \alpha \geq 1$ ), the general case remaining conjectured.
- **Honesty** (theorem 4.14). An *eikonal* price — each nat of knowledge costing the same length everywhere — is equivalent to the proportionality  $U = cJ$ : the Fisher family  $\mathcal{F}$ .
- **Stam rigidity** (theorems 4.17 and 4.18). The honest geometry is *hyperbolic*, with curvature  $K = -1/(2cJ_0)$  on every location-scale family, and the Stam bound ( $J_0 \geq 1$ , equality iff Gaussian) makes the *Gaussian* the most hyperbolic belief — the value  $-1/4$  being its image under a unit (theorem 4.21).

**A common honesty guarantee.** These three results share one and the same honesty guarantee, made explicit by a statement closing Section 4.4: the *gauge invariance of cost* (theorem 4.21). Changing the cost unit —  $(e, U) \mapsto (\lambda e, \lambda U)$  — acts as a homothety:  $\tilde{g} \mapsto \lambda \tilde{g}$ ,  $d \mapsto \sqrt{\lambda} d$ ,  $K \mapsto K/\lambda$ . The wall, the eikonal  $U = cJ$ , and the ranking of curvatures (the extremal Gaussian) are its invariants; the value  $-1/4$  is an image.

On this class, inference is *well-posed*: the minimizer exists and does not collapse toward the boundary, and the cost of reaching a given precision has a floor, intrinsic to the geometry, that diverges as one approaches certainty. *Throughout the article, “well-posed” is meant in the sense of existence and stability, uniqueness remaining open* (Sections 5 and 6). Thermodynamics *motivates*



**Figure 2: The cost geometry of belief.** (a) The location-scale leaf in the *Poincaré disk*: the geodesics (arcs perpendicular to the boundary); the highlighted arc joins two beliefs  $A$  and  $B$ , the straight chord dashed), the learning trajectory  $n = 1, 5, 20, 60$  toward the boundary, and the boundary  $\sigma=0$  (the Diracs) at infinite distance — the *wall*; curvature  $K = -1/(2cJ_0)$  ( $-1/(4J_0)$  at the representative  $c=2$ , a unit image, theorem 4.21). (b) The same leaf as a Gaussian *pseudosphere* (local model, Hilbert’s theorem (do Carmo, 1976)): the width equals the spread  $\sigma$  (iso-entropy circles), and the descent  $n = 1, \dots, 60$  reaches the cusp ( $H \rightarrow -\infty$ ); the Gaussian is its extremal leaf (Stam,  $J_0=1$ ). All quantities are verified numerically.

this framework; the proofs are carried out in metric geometry, from standard borrowed tools (Section 4.1), whose pivot is the Otto identity  $\|\nabla_{\mathcal{W}_2} H\|^2 = J$ .

Geometrically, these three results can be read on a single image. figure 2 gives the intuition — the inference chain that produces the belief, from sensor to posterior, being detailed in figure 1. In (a), on the location-scale leaf, it is exactly the hyperbolic half-plane: the Fisher information  $J = 1/\sigma^2$  diverges as the belief tightens, the cost metric  $g = \frac{2cJ_0}{\sigma^2} \mathbf{I}$  stretches, and the boundary — certainty, a Dirac — is pushed to infinite distance; this *wall* makes inference well-posed (an interior minimizer exists). The curvature  $K = -1/(2cJ_0)$  there *reads* the Fisher information without creating it. The geodesics there are the paths of *minimal cost* between beliefs: a vertical changes only the confidence, whereas changing one’s mind (the arc from  $A$  to  $B$ ) passes through a more *diffuse* belief — one must open up to move at lesser cost, where the straight chord, at frozen confidence, costs more. In (b), the same leaf as a pseudosphere, the width equaling the spread  $\sigma$  (iso-entropy); the Gaussian is, on location-scale families, its most curved leaf (Stam, 1959). The “bridge” between the inference chain (figure 1) and this geometry is the Otto identity  $\|\nabla_{\mathcal{W}_2} H\|^2 = J$ , a transport fact.

**Plan.** Section 2 poses the question and the formal framework; we then characterize the geometry (Section 4) — the wall (Section 4.3), the eikonal (Section 4.4), hyperbolicity and Stam rigidity (Section 4.4); we draw from it a well-posed inference (Section 5); we discuss scope and limits (Section 6), and we situate the work (Section 7). The machinery proofs are deferred to the appendix (Appendix A).

## 4 Characterizations

We first gather the borrowed tools (Section 4.1) and the machinery valid for every candidate (Section 4.2), then establish the three characterizations: the *boundary* (the wall), the *honest family*

**Box 1** (Algorithmic application). Three objects of machine learning fall into this framework — two reduce to it exactly, one shares its rule (Fisher = cost of change) —, separated by  $\sigma$ : inside ( $\sigma > 0$ ) a belief is revised at finite cost; at the boundary ( $\sigma = 0$ , the wall  $\partial_\infty$ ) one must retrain.

**1. Kalman — interior point.** In the linear-Gaussian regime, the belief of a scalar Kalman filter (Kalman, 1960) is the *exact* posterior  $b_t = \mathcal{N}(\mu_t, \sigma_t^2)$ , a point of the Gaussian leaf (figure 2 (a)); its recurrence moves within it — prediction *opens* the belief, correction *tightens* it. This point is at finite cost distance from any other interior belief (corollary 5.2), hence revisable at finite cost by each datum.

**2. The point predictor — the wall.** A point estimate  $\hat{x}$  (variance-free regression, arg max, collapsed output) is modeled as the Dirac  $\delta_{\hat{x}}$ , a point of  $\partial_\infty$  ( $H = -\infty$ ). By theorem 4.10(i), it is at *infinite* cost distance from any interior point: no continuous revision exists, one can only *retrain*. The overconfident softmax is its practical instance, *near* the wall ( $H \rightarrow -\infty$  without reaching it).

**3. EWC — the same rule, carried to weights.** *Elastic Weight Consolidation* (Kirkpatrick et al., 2017) penalizes the displacement of weights by  $\sum_i F_i (\theta_i - \theta_i^*)^2$ ,  $F_i$  the diagonal of the Fisher information: the cost of changing a parameter is proportional to its *certainty* — the same principle as the cost metric, carried from beliefs to parameters. *Scope*: this  $F_i$  is a Fisher–Rao tensor on the weights (remark 2.5), whereas the  $J$  of the framework is the slope of  $-H$  on transport — a kinship of principle, not a geometric identity.

(the eikonal  $U = cJ$ ), and the *curvature* (hyperbolicity and Stam rigidity).

#### 4.1 Borrowed tools

**Assumption 4 (B1 — the Otto identity).** On  $(\mathcal{P}_2, \mathcal{W}_2)$ ,  $-H$  is geodesically convex (McCann, 1997) and  $\|\nabla_{\mathcal{W}_2} H\|^2 = J$  — the Otto identity (Otto, 2001), here via Ambrosio et al. (2008, §10.4). *This is the crux of the argument, the converter “uniform price”  $\rightarrow$  “cost  $\propto J$ ”.*

**Assumption 5 (B2 — strong upper gradient).** For every  $AC$  curve:  $|\text{Ent}(\gamma_t) - \text{Ent}(\gamma_s)| \leq \int_s^t \sqrt{J} |\dot{\gamma}| dr$ , the left-hand side finite and  $\text{Ent} \circ \gamma$  absolutely continuous as soon as the right-hand side is (Ambrosio et al., 2008, §§1.2, 2.4).

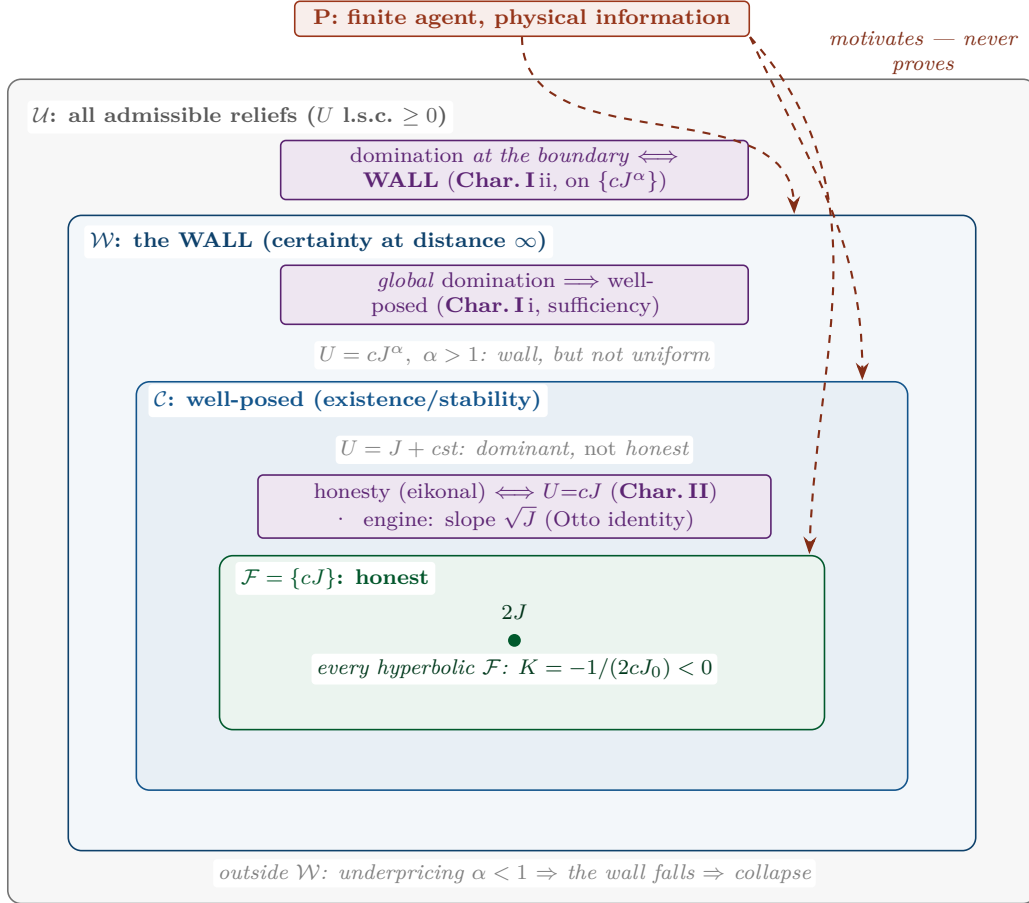
**Assumption 6 (B3 — compatibility  $(\mathcal{W}_2, \tau)$ ).**  $(\mathcal{P}_2, \mathcal{W}_2)$  Polish and geodesic (Ambrosio et al., 2008, ch. 7);  $\mathcal{W}_2$  is  $\tau$ -l.s.c.; the  $\{m_2 \leq M\}$  are  $\tau$ -compact.

**Assumption 7 (B4 — flat Gaussian leaf).**  $\mathcal{W}_2^2(\mathcal{N}(\mu_1, \sigma_1^2), \mathcal{N}(\mu_2, \sigma_2^2)) = (\mu_1 - \mu_2)^2 + (\sigma_1 - \sigma_2)^2$  (Givens and Shortt, 1984): the leaf  $\mathcal{G} = \{\mathcal{N}(\mu, \sigma^2)\}$  is isometric to the Euclidean half-plane  $\{\sigma > 0\}$ ; McCann’s geodesics stay Gaussian there (Ambrosio et al., 2008, ex. 7.3.14).

**Assumption 8 (B5 — metric Hopf–Rinow).** A locally compact length space in which bounded closed balls are compact is complete and geodesic (Burago et al., 2001, thm 2.5.28).

**Assumption 9 (B6 — l.s.c. dual of Fisher).**  $J(q) = \sup_{\psi \in C_c^\infty} \{-2 \int \Delta \psi dq - \int |\nabla \psi|^2 dq\}$  for every  $q \in \mathcal{P}(\mathbb{R}^n)$  (Dupuis and Ellis, 1997, lem. D.45); as a supremum of narrowly continuous functionals,  $J$  is narrowly l.s.c. (Hence  $J \in \mathcal{U}$  and, for every  $c > 0$ ,  $cJ \in \mathcal{U}$ .)

**Assumption 10 (B7 — flatness of 1D location-scale families).** The quantile map  $p \mapsto F_p^{-1}$  is an isometry of  $(\mathcal{P}_2(\mathbb{R}), \mathcal{W}_2)$  onto the convex cone of inverse cumulative distribution functions in  $L^2(0, 1)$  (Santambrogio, 2015; Bobkov and Ledoux, 2019). For a base  $\varphi_0$  (density, mean 0, variance 1, quantile  $Q_0$ ), the location-scale family  $\mathcal{L}_{\varphi_0} = \{p_{\mu, \sigma} = \sigma^{-1} \varphi_0((\cdot - \mu)/\sigma)\}$  is there the affine 2-plane  $\{\mu \mathbf{1} + \sigma Q_0\}$ , with  $\langle \mathbf{1}, \mathbf{1} \rangle = 1$ ,  $\langle \mathbf{1}, Q_0 \rangle = 0$ ,  $\langle Q_0, Q_0 \rangle = 1$ :  $(\mathcal{L}_{\varphi_0}, \mathcal{W}_2)$  is thus *isometric to the*



**Figure 3: The universe of cost geometries.** The nested rings  $\mathcal{F} \subset \mathcal{C} \subset \mathcal{W} \subset \mathcal{U}$ : the wall  $\mathcal{W}$  (theorem 4.10), the well-posed class  $\mathcal{C}$ , the honest Fisher family  $\mathcal{F}$  (theorem 4.14), and at their center the Gaussian, the most hyperbolic belief (Stam rigidity, theorem 4.18). The physics **P** motivates the boundaries (dashed) without entering any proof (proposition B.1).

Euclidean half-plane  $\{\sigma > 0\}$ , with metric  $d\mu^2 + d\sigma^2$ . (Generalizes assumption 7: the Gaussian is the case  $\varphi_0 = \mathcal{N}(0, 1)$ .)

Metric setting.  $\gamma : [0, T] \rightarrow \mathcal{P}_2$  is  $AC^2$  if there exists  $m \in L^2$  with  $\mathcal{W}_2(\gamma_s, \gamma_t) \leq \int_s^t m$ ;  $|\dot{\gamma}|$  exists a.e. (Ambrosio et al., 2008, thm 1.1.2). We set  $K(\gamma) := \sup_{\text{part.}} \sum_i \mathcal{W}_2^2(\gamma_{t_i}, \gamma_{t_{i+1}})/(t_{i+1} - t_i)$ .

**Lemma 4.1** (kinetic action). (a)  $\gamma \in AC^2 \implies K(\gamma) = \int_0^T |\dot{\gamma}|^2$ . (b)  $K(\gamma) < \infty \implies \gamma \in AC^2$ . (c)  $K$  is  $\tau$ -pointwise l.s.c.

*Proof.* (a) ( $\leq$ , i.e.  $K \leq \int |\dot{\gamma}|^2$ ) Cauchy–Schwarz on each subinterval,  $\mathcal{W}_2^2(\gamma_{t_i}, \gamma_{t_{i+1}}) \leq (t_{i+1} - t_i) \int_{t_i}^{t_{i+1}} |\dot{\gamma}|^2$ , sum and then take the sup. ( $\geq$ , i.e.  $\int |\dot{\gamma}|^2 \leq K$ ) the dyadic quotients  $f_N \rightarrow |\dot{\gamma}|$  a.e. (a.e. limit, (Ambrosio et al., 2008, thm 1.1.2), not the Cauchy–Schwarz bound); Fatou:  $\int |\dot{\gamma}|^2 \leq \liminf \int f_N^2 \leq K$ . (b) the inequality  $\frac{(x+y)^2}{\alpha+\beta} \leq \frac{x^2}{\alpha} + \frac{y^2}{\beta}$  yields additivity of  $K$ , hence absolute continuity and then  $\int |\dot{\gamma}|^2 \leq K$ . (c) each term is  $\tau$ -l.s.c. (assumption 6); a sup of l.s.c. is l.s.c.  $\square$

**Lemma 4.2** (entropy, slope — recall of imports). With  $H, J$  in the sense of definition 2.1: Ent  $> -\infty$  on  $\mathcal{P}_2$ , geodesically convex (McCann, 1997); its metric slope equals  $\sqrt{J}$  (assumption 4); it is a strong upper gradient (assumption 5).

## 4.2 What holds for every candidate: existence and geodesics

This entire section holds for *every*  $U \in \mathcal{U}$ : it is the common machinery of the whole class, independent of the choice of relief. For  $\gamma \in AC^2$  joining  $p_0$  to  $p_1$ :  $\mathcal{A}_T(\gamma) := \int_0^T [\frac{1}{2}|\dot{\gamma}|^2 + U(\gamma_t)]dt$ ,  $\Phi(T) := \inf \mathcal{A}_T$ ,  $\Psi(e) := \inf \ell_e = d_e$ . The statements below hold for every candidate; their proofs, with their regime clauses, are collected in Appendix A.

**Lemma 4.3** (AM–GM).  $\forall a, b \geq 0$ :  $\frac{1}{2}a^2 + b \geq \sqrt{2b}a$ , with equality iff  $a = \sqrt{2b}$ .

This inequality converts the action  $\frac{1}{2}|\dot{\gamma}|^2 + (e + U)$  into length  $\sqrt{2(e + U)}|\dot{\gamma}|$ , with saturation at the Jacobi speed.

**Theorem 4.4** (Maupertuis–Jacobi correspondence (Arnold, 1989)).  $\forall U \in \mathcal{U} \forall e > 0$ , writing  $p_0, p_1 \in \{U < \infty\}$ : (i)  $\forall \gamma \in AC^2$ :  $\mathcal{A}_T(\gamma) \geq \ell_e(\gamma) - eT \geq \Psi(e) - eT$ ; (ii) if  $\hat{\gamma}$  minimizes  $\ell_e$  ( $\Psi(e) < \infty$ ,  $U(\hat{\gamma}) < \infty$  a.e.), its constant-energy reparametrization  $|\dot{\gamma}| = \sqrt{2(e + U)}$  exists, with parameter  $T_e = \int |\dot{\gamma}|/\sqrt{2(e + U)}$ , and minimizes  $\mathcal{A}_{T_e}$  with  $\Phi(T_e) = \Psi(e) - eT_e$  (the bound  $L/\sqrt{2e}$  requires  $e > 0$ : this is what makes the reparametrization licit); (iii) if  $\gamma^*$  minimizes  $\mathcal{A}_T$  and saturates (i), then  $\frac{1}{2}|\dot{\gamma}^*|^2 = e + U(\gamma^*)$  a.e.

**Theorem 4.5** (existence of geodesics).  $\forall U \in \mathcal{U}$ , if  $\Phi(T) < \infty$ , then  $\mathcal{A}_T$  admits a minimizer in  $AC^2([0, T]; \mathcal{P}_2)$  joining  $p_0$  to  $p_1$ . (Same for  $\ell_e$ ,  $e > 0$ , (Buttazzo et al., 1998, §4.3).)

**Proposition 4.6** (energy law).  $\forall U \in \mathcal{U}$ , every minimizer  $\gamma^*$  of  $\mathcal{A}_T$  (provided by theorem 4.5,  $\mathcal{A}_T(\gamma^*) < \infty$ ) satisfies  $\frac{1}{2}|\dot{\gamma}^*|^2 = U(\gamma^*) + e$  a.e., for some constant  $e \in \mathbb{R}$ .

The constant  $e \in \mathbb{R}$  thus obtained is, when it is  $> 0$ , the *level* of definition 2.6 (which requires  $e \geq 0$ ); its useful regime  $e > 0$  is discussed in theorem 4.7.

**Theorem 4.7** (optimal curve  $\Leftrightarrow$  geodesic).  $\forall U \in \mathcal{U}$ , let  $\gamma^*$  be a minimizer of  $\mathcal{A}_T$  and  $e \in \mathbb{R}$  its constant (proposition 4.6); in the regime  $e > 0$  (equivalently:  $T$  small enough;  $e > 0$  may fail for large  $T$  — cusps), the only regime where  $\ell_e$  is a distance:  $\gamma^*$  traverses its trace at the Jacobi speed, and under exact duality  $\Phi(T) = \sup_{x>0} \{\Psi(x) - xT\}$  its trace is a geodesic of  $\tilde{g}_{e,U}$  in the length sense (of  $d_e = \inf \ell_e$ ;  $\tilde{g}$  is not a smooth Riemannian metric for  $U$  merely l.s.c.). The converse is unconditional. (Exact duality is not automatic: it can fail through a first-order transition — a two-route counterexample (one short but costly, the other long but free) exhibits a jump in the energy.)

## 4.3 Characterization I: the wall

We first characterize the wall. A first equivalence shows that pushing certainty to infinite distance — the condition for a well-posed inference (definition 4.8) — amounts to dominating the Fisher information. The map of admissible classes (figure 3) locates this wall: the ring  $\mathcal{W}$  of reliefs that raise the boundary to infinite distance.

First equivalence. The operational constraint of remark 2.3(b) entails that the relief *dominates* the Fisher information *at the boundary*: on the power family, this is the *necessary* condition for the wall (theorem 4.10(ii)). *Global* domination, strictly stronger, is its unconditional *sufficient* form (theorem 4.10(i)); it defines the well-posed class (definition 4.8).

**Definition 4.8** (well-posed class).

$$\mathcal{C} := \{U \in \mathcal{U} \mid \exists \varepsilon > 0, U \geq \varepsilon J \text{ pointwise on } \mathcal{P}_2\} \subset \mathcal{U}$$

(*global* domination: one same  $\varepsilon$  at every point).

**Definition 4.9** (wall class).  $\mathcal{W} := \{U \in \mathcal{U} : \partial_\infty \text{ is at infinite } d_e\text{-distance}\}$  — the reliefs that *raise the wall*. Global domination entails it (theorem 4.10(i)), hence  $\mathcal{C} \subset \mathcal{W}$ , strictly (on the family,  $U = cJ^\alpha$  with  $\alpha \in (1, \infty)$  raises the wall without belonging to  $\mathcal{C}$ ). Whence the chain  $\mathcal{F} \subset \mathcal{C} \subset \mathcal{W} \subset \mathcal{U}$ .

**Theorem 4.10** (characterization of the wall). (i) Sufficiency (every  $U \in \mathcal{C}$ : the wall).

$$U \geq \varepsilon J \implies \forall e > 0 \forall \gamma \in AC^2 : \ell_e(\gamma) \geq \sqrt{2\varepsilon} |H(p_1) - H(p_0)|;$$

$-H$  is thus  $\frac{1}{\sqrt{2\varepsilon}}$ -Lipschitz for  $d_e$ , and the wall holds:  $\partial_\infty = \{H = -\infty\}$  is at infinite  $d_e$ -distance (every dimension), whereas it is at finite  $\mathcal{W}_2$  distance. Since the wall pushes  $\partial_\infty$  to infinity, no minimizing sequence of finite cost collapses onto it: the minimizer of the action (theorem 4.5) is interior — with fixed endpoints, or for estimation as soon as the fidelity is coercive; the general free-boundary case (mass escape) falls under remark 6.3. (ii) Necessity, on the family  $U = cJ^\alpha$  ( $c > 0, \alpha \in \mathbb{R}$ ). On the Gaussian leaf,

$$[\text{wall: } \partial_\infty \text{ at infinite } d_e\text{-distance}] \iff \alpha \geq 1 \quad (\text{domination at the boundary } J \rightarrow \infty).$$

$\alpha < 1$  brings  $\partial_\infty$  back to finite distance. (Calibration.) Global domination (membership in  $\mathcal{C}$ ) is strictly stronger: on this family,  $cJ^{\alpha-1} \geq \varepsilon$  for every  $J \in (0, \infty)$  forces  $\alpha = 1$ . Thus  $\alpha \in (1, \infty)$  has the wall (boundary domination) without belonging to  $\mathcal{C}$ , which on the family coincides with  $\alpha = 1$  (theorem 4.14). (iii) Scope. Sufficiency (i) is unconditional (every  $U \in \mathcal{C}$ ); the general necessity (every  $U \in \mathcal{U}$ : “no boundary domination  $\Rightarrow$  no wall”) is conjectured (remark 6.1).

*Proof.* (i)  $U \geq \varepsilon J \implies 2(e + U) \geq 2\varepsilon J$ , whence  $\sqrt{2(e+U)} \geq \sqrt{2\varepsilon}\sqrt{J}$ ; by lemma 4.2,  $\ell_e \geq \sqrt{2\varepsilon} \int \sqrt{J} |\dot{\gamma}| \geq \sqrt{2\varepsilon} |\Delta \text{Ent}|$ . If  $\text{Ent}(p_1) = +\infty$ , a finite joining length is impossible (otherwise the strong upper gradient would make  $\text{Ent} \circ \gamma$  a.c., hence finite at  $t = 1$ ; or, by narrow l.s.c. of  $\text{Ent}$ , would force  $\text{Ent}(p_1) \leq \liminf < \infty$ ):  $\ell_e = +\infty$ . Interiority of the action minimizer follows from theorem 4.5. (ii) On  $\mathcal{G}$  (assumption 7),  $J = 1/\sigma^2$ ,  $U = c\sigma^{-2\alpha}$ . Along the ray  $\mu = \text{const}$ ,  $\ell_e = \int \sqrt{2(e+U)} d\sigma$  exactly (witness path, for convergence); and  $|\dot{\gamma}| \geq |\dot{\sigma}|$  gives, for any curve, the lower bound  $\ell_e \geq \int \sqrt{2(e+U)} |\dot{\sigma}|$ . At  $e = 0$ , near  $\sigma = 0$ ,  $\sqrt{2U} d\sigma = \sqrt{2c} \sigma^{-\alpha} d\sigma$  and  $\int_0^{\sigma_0} \sigma^{-\alpha} d\sigma < \infty \iff \alpha < 1$ : thus  $\alpha \geq 1$  makes the radial lower bound diverge (wall) and  $\alpha < 1$  gives the witness path a finite length (boundary at finite distance). The case  $e > 0$ :  $\sqrt{2U} \leq \sqrt{2(e+U)} \leq \sqrt{2e} + \sqrt{2U}$  preserves divergence ( $\alpha \geq 1$ , via the lower bound  $\sqrt{2U}$ ) and convergence ( $\alpha < 1$ , via the upper bound  $\sqrt{2e} + \sqrt{2U}$ , integrable on  $(0, \sigma_0]$ ). (iii) cf. remark 6.1.  $\square$

**Corollary 4.11** (the wall, answer to the problem of finiteness). Under remark 2.3 (boundary excluded + tenable exit), a conformal cost geometry is well-posed for existence and stability only if it raises the wall, hence only if its relief dominates  $J$  at the boundary (on the power family  $\{cJ^\alpha\}$ ,  $\alpha \geq 1$ ; theorem 4.10(ii)). On the power family, the wall is the condition that finiteness requires (open boundary excluded  $\rightarrow$  collapse). The class  $\mathcal{C}$  (global domination) is its convenient uniform form; on the family it restricts to  $\alpha = 1$ , in agreement with theorem 4.14.

#### 4.4 Characterization II: eikonal $\Leftrightarrow$ Fisher family

A second equivalence refines the well-posed class. Requiring an eikonal price — the same length per nat at every point — selects exactly the reliefs proportional to Fisher information, the family  $\mathcal{F} = \{cJ\}$ . This is a uniqueness of *family*, not of object.

**Definition 4.12** (honesty of the price — eikonal).  $U \in \mathcal{C}$  is *eikonal* if knowledge  $-H$  has a constant metric slope for  $\tilde{g}_{0,U}$ :  $\exists \kappa > 0$  such that  $\|\nabla_{\tilde{g}_{0,U}} H\| = \kappa$  on  $\{0 < U < \infty\}$  (where  $\|\nabla_{\mathcal{W}_2} H\| = \sqrt{J}$ , assumption 4).

**Definition 4.13** (Fisher family).  $\mathcal{F} := \{U = cJ : c > 0\} \subset \mathcal{C}$ : the reliefs proportional to Fisher information (exponent  $\alpha = 1$ ;  $\mathcal{F} \subset \mathcal{C}$  by proposition 4.23).

**Theorem 4.14** (characterization of the eikonal family). For every  $U \in \mathcal{C}$ :

$$U \text{ eikonal} \iff \exists c > 0, U = cJ \text{ pointwise on } \{0 < U < \infty\},$$

and then  $\kappa = 1/\sqrt{2c}$ . Eikonality of the price thus selects exactly the Fisher family  $\mathcal{F} = \{cJ : c > 0\}$  (definition 4.13). (Corollary: on power laws  $U = cJ^\alpha$ , eikonal  $\Leftrightarrow \alpha = 1$ .)

*Proof.* Conformal contraction (assumption 4): for  $\tilde{g}_{0,U} = 2U g_{\mathcal{W}_2}$  and  $\|\nabla_{\mathcal{W}_2} H\| = \sqrt{J}$ , we have  $\|\nabla_{\tilde{g}_{0,U}} H\| = \sqrt{J}/\sqrt{2U}$ . This slope equals a constant  $\kappa > 0$  at every point if and only if  $\sqrt{J}/\sqrt{2U} = \kappa$ , i.e.  $U = J/(2\kappa^2) = cJ$  with  $c = 1/(2\kappa^2)$ ; conversely  $U = cJ$  gives  $\kappa = 1/\sqrt{2c}$ . (The restriction to power laws is unnecessary: the iff is pointwise.)  $\square$

*Remark 4.15* (uniqueness of family, not of object). theorem 4.14 is a *uniqueness of proportionality* ( $U \propto J$ , family  $\mathcal{F} = \{cJ\}$ ), not of the constant: the *whole* family is eikonal, its members differing only by the unit  $\kappa = 1/\sqrt{2c}$  (the “price of a nat”). We thus claim no unique object; we have characterized a *family*.

*Remark 4.16* (honesty of the price). The eikonal condition reads as an honesty of pricing: each nat costs the same length everywhere. Under this reading, the eikonal aligns with Landauer’s principle (Landauer, 1961; Bérut et al., 2012) — each nat dissipating everywhere the same energy  $k_B T$  — as a single non-distortion measured in two units, length and energy. The bridge between the two is the Otto identity  $\|\nabla_{\mathcal{W}_2} H\|^2 = J$  (assumption 4), a transport fact that turns “uniform price” into proportionality to  $J$ . This identification reads off an already-established geometry (theorem 4.14); it enters no proof, and all theorems are proved in nats. At the boundary, distance and energy diverge together (corollary 5.3). Quantitatively, at  $e = 0$ ,  $d_0 = \sqrt{2c} |\Delta H| = (\sqrt{2c}/k_B T) E$ : the geometric floor and Landauer’s energy rise together.

The eikonal cost geometry is hyperbolic, and its curvature reads the Fisher information of the base. On every location-scale family it equals  $K = -1/(2cJ_0)$ , and the Stam bound makes the Gaussian extremal — the most hyperbolic location-scale belief. What is unit-invariant — the *invariant content* — is the sign, the *ranking* of the leaves by curvature and its extremum (the Gaussian); the value  $-\frac{1}{4}$  is its image under the unit  $c = 2$  (remark 4.20).<sup>1</sup>

**Theorem 4.17** (universal hyperbolicity of  $\mathcal{F}$ ).  $\forall U = cJ \in \mathcal{F} \forall e \geq 0$ , on the Gaussian leaf  $(\mathcal{G}, \tilde{g}_{e,U})$ , the Gauss curvature equals, in closed form,

$$K(\mu, \sigma) = -\frac{c(c + 3e\sigma^2)}{2(c + e\sigma^2)^3} < 0 \quad (\forall \sigma > 0).$$

The whole Fisher family is thus strictly hyperbolic. At  $e = 0$ :  $K \equiv -1/(2c)$  for all  $\sigma$  (exact Poincaré half-plane (Burago et al., 2001)). For  $e > 0$ :  $K$  increases from  $-1/(2c)$  ( $\sigma \rightarrow 0$ ) to 0 ( $\sigma \rightarrow \infty$ ) — monotone,  $\partial_\sigma K = 6ce^2\sigma^3/(c + e\sigma^2)^4 > 0$ . The ranking of the leaves by curvature is defined at  $e = 0$ , where  $K$  is constant; this is the setting of Stam rigidity (theorem 4.18) and of theorem 4.21.

*Proof.*  $\tilde{g}_{e,U} = f(d\mu^2 + d\sigma^2)$ ,  $f = 2e + 2c/\sigma^2$ ; for a conformal metric of the plane,  $K = -\Delta(\ln f)/(2f)$  ( $\Delta$  Euclidean,  $f$  depending only on  $\sigma$ ). Direct computation:

$$\Delta(\ln f) = \partial_\sigma^2(\ln f) = \frac{2c(c + 3e\sigma^2)}{\sigma^2(c + e\sigma^2)^2}, \quad 2f = \frac{4(c + e\sigma^2)}{\sigma^2},$$

whence  $K = -\frac{2c(c + 3e\sigma^2)/(\sigma^2(c + e\sigma^2)^2)}{4(c + e\sigma^2)/\sigma^2} = -\frac{c(c + 3e\sigma^2)}{2(c + e\sigma^2)^3}$ . Numerator and denominator strictly positive ( $c > 0$ ,  $e \geq 0$ ,  $\sigma > 0$ ):  $K < 0$  everywhere. The limits and monotonicity follow by direct reading.  $\square$

**Theorem 4.18** (Stam rigidity — the Gaussian, the most hyperbolic location-scale belief). Let  $\varphi_0$  be a standardized regular base ( $\int x\varphi_0 = 0$ ,  $\int x^2\varphi_0 = 1$ ,  $J_0 := J(\varphi_0) < \infty$ ) and  $\mathcal{L}_{\varphi_0}$  the location-scale family it generates (assumption 10). For the representative  $U = 2J$ , at  $e = 0$ :

- (i)  $(\mathcal{L}_{\varphi_0}, \mathcal{W}_2)$  is isometric to the Euclidean half-plane (assumption 10), and  $J(p_{\mu,\sigma}) = J_0/\sigma^2$ , whence  $U = 2J_0/\sigma^2$ ;

<sup>1</sup>The two closed forms ( $K < 0$ , and  $K = -1/(2cJ_0)$  on the location-scale families) are confirmed symbolically (sympy); scripts and outputs in the verification companion (cf. findings).

(ii) the intrinsic Gauss curvature of  $(\mathcal{L}_{\varphi_0}, \tilde{g}_{0,2J})$  is constant,

$$\boxed{K_{\varphi_0} = -\frac{1}{4J_0}} < 0;$$

(iii) (Stam extremum)  $J_0 \geq 1$  (Stam, 1959), with equality if and only if  $\varphi_0$  is Gaussian (assumption 9,  $\text{Var} = 1$ ); hence  $K_{\varphi_0} \in [-\frac{1}{4}, 0)$ , and the extremal value  $-\frac{1}{4}$  is attained by the Gaussian leaf (gauge  $c = 2$ ,  $\text{Var}(\varphi_0) = 1$ ; remark 4.22). (Scope outside location-scale: remark 6.2.)

*Proof.* (i) Flatness: assumption 10. Scaling:  $J(p_{\mu,\sigma}) = \int (p'_{\mu,\sigma})^2 / p_{\mu,\sigma} = \sigma^{-2} \int (\varphi'_0)^2 / \varphi_0 = J_0 / \sigma^2$ . (ii) On  $\mathcal{L}_{\varphi_0}$ ,  $g_{\mathcal{W}_2} = d\mu^2 + d\sigma^2$  (assumption 10) and  $\tilde{g}_{0,2J} = f(d\mu^2 + d\sigma^2)$  with  $f = 2U = 4J_0 / \sigma^2$ ;  $\Delta(\ln f) = 2/\sigma^2$ ,  $2f = 8J_0 / \sigma^2$ , whence  $K = -(2/\sigma^2) / (8J_0 / \sigma^2) = -1/(4J_0)$ . (iii) Cramér–Rao for a density of variance 1:  $J_0 \text{Var} = J_0 \geq 1$ , with equality iff the score  $\varphi'_0 / \varphi_0$  is affine, i.e. iff  $\varphi_0$  is Gaussian. Whence  $-1/(4J_0) \in [-\frac{1}{4}, 0)$ , extremal at the Gaussian. (“Stam” in the broad sense: the bound used is the Gaussian extremality of information under fixed variance.)  $\square$

*Remark 4.19* (why position information, and not scale). What distinguishes this rigidity is the choice of the quantity that drives the curvature. The conformal cost charges the slope of  $-H$  along transport (the Otto identity,  $\|\nabla_{\mathcal{W}_2} H\|^2 = J$ , assumption 4), and  $J$  is the position Fisher information:  $J(p_{\mu,\sigma}) = J_0 / \sigma^2$  with  $J_0 = \int (\varphi'_0)^2 / \varphi_0$ . This is the quantity that Cramér–Rao/Stam bounds below ( $J_0 \geq 1$ , variance 1). The Fisher–Rao geometry of location-scale families (symmetric base) is also hyperbolic of constant curvature, but  $\propto -1/b_0^2$  with  $b_0^2$  the scale information (Nielsen, 2021) — which Stam does not bound: hence the absence of this extremum in Fisher–Rao. The conformal cost of transport makes position information carry the curvature, which is what makes Stam rigidity possible (remark 7.1). Moreover,  $K = -1/(4J_0)$  holds for every standardized base — assumption 10 gives  $g_{\mathcal{W}_2}$  flat without a symmetry hypothesis —, whereas the Fisher–Rao comparison requires a symmetric base: cost rigidity thus covers a larger class of bases.

*Remark 4.20* (the honest representative  $U = 2J$ ). Within  $\mathcal{F}$ , we choose the representative  $c = 2$ , that is

$$\boxed{U = 2J}, \quad \tilde{g}_e = 2(e + 2J) g_{\mathcal{W}_2}.$$

This choice is a unit normalization (one nat = two lengths: at  $e = 0$ ,  $\kappa = 1/2$ , whence  $d = 2|\Delta H|$ , the exact eikonal) that also fixes the curvature to  $-\frac{1}{4}$  on the Gaussian leaf (theorem 4.17,  $c = 2$ ,  $J_0 = 1$ ). The load-bearing results (theorem 4.10: wall, well-posed; theorem 4.17, theorem 4.18: hyperbolicity,  $K = -1/(2cJ_0)$ ) hold for the whole class / family. Changing the representative is a change of unit (theorem 4.21): it rescales the value, the distances and the curvature, and preserves the sign, the ranking and the Gaussian extremum.

*Change of cost unit.* We treat the level  $e$  and the relief  $U$  as the same kind of quantity — cost densities —, so that a change of unit rescales them together: for  $\lambda > 0$ ,  $(e, U) \mapsto (\lambda e, \lambda U)$  (on the family  $\mathcal{F} = \{cJ\}$ ,  $c \mapsto \lambda c$  and  $e \mapsto \lambda e$ ).

**Theorem 4.21** (gauge invariance of cost). *Under the change of cost unit  $(e, U) \mapsto (\lambda e, \lambda U)$ ,  $\lambda > 0$ , the cost geometry (2.1) transforms by homothety, for all  $e \geq 0$  and all  $U \in \mathcal{C}$ :*

$$\tilde{g}_{e,U} \mapsto \lambda \tilde{g}_{e,U}, \quad d_e \mapsto \sqrt{\lambda} d_e, \quad K \mapsto K/\lambda$$

(the last on the location-scale leaves, of dimension 2). Thus gauge-invariant: the sign of  $K$  and hyperbolicity; the finiteness or infiniteness of the  $d_e$ -distances — the wall (theorem 4.10) — and well-posedness; the traces of the geodesics and the ratios of distances; and the eikonal proportionality  $U = cJ$  (theorem 4.14). At  $e = 0$ , where  $K$  is constant (theorem 4.17), one further gains the ranking of the leaves by curvature and its Gaussian extremum (theorem 4.18), whose ratio  $K(\text{gauss})/K(\varphi_0) = J_0$ . Depending on the unit (scale images): every absolute distance, the price of a nat  $\kappa = 1/\sqrt{2c}$ , and the value  $K = -1/(2cJ_0)$  (i.e.  $-1/(4J_0)$  at the representative  $c = 2$ , and  $-\frac{1}{4}$  for the Gaussian  $J_0 = 1$ ).

*Proof.* Under  $(e, U) \mapsto (\lambda e, \lambda U)$ ,  $\tilde{g}_{e,U} = 2(e + U)g_{\mathcal{W}_2}$  becomes  $2(\lambda e + \lambda U)g_{\mathcal{W}_2} = \lambda \tilde{g}_{e,U}$  (all  $e \geq 0$ ). A length  $\ell_e = \int \sqrt{2(e + U)} |\dot{\gamma}|$  is thus multiplied by  $\sqrt{\lambda}$ , whence  $d_e = \inf \ell_e \mapsto \sqrt{\lambda} d_e$ ; the infimum being over the same set of curves, the minimizing traces, the ratios of distances and the finiteness/infiniteness of the lengths — hence the wall (theorem 4.10) — are unchanged. On a conformal leaf  $\tilde{g} = f(d\mu^2 + d\sigma^2)$ ,  $K = -\Delta(\ln f)/(2f)$ , and  $f \mapsto \lambda f$  gives  $\Delta \ln(\lambda f) = \Delta \ln f$  with a denominator  $2\lambda f$ , whence  $K \mapsto K/\lambda$  pointwise. A positive scalar factor preserves the sign and curvature ratios; at  $e = 0$ , where  $K$  is constant (theorem 4.17), it moreover preserves the *ranking* of the leaves, whence the Gaussian extremum (theorem 4.18). Finally the eikonal slope  $\|\nabla_{\tilde{g}_{0,U}} H\| = \sqrt{J}/\sqrt{2U} = \kappa$  becomes  $\kappa/\sqrt{\lambda}$ , a nonzero constant: the proportionality  $U = cJ$  (theorem 4.14) is preserved (its value  $\kappa$ , in turn, is gauge).  $\square$

*Remark 4.22* (invariants and choice of unit). The wall, the eikonal and Stam rigidity are the invariants of this change of unit, and no proven conclusion depends on an absolute cost. The value  $-\frac{1}{4}$  rests on two conventions: the gauge  $c = 2$  and the standardization  $\text{Var}(\varphi_0) = 1$ , which gives  $J_0 = 1$ ; the invariant content is the sign, the ranking, the extremum and the ratio  $K(\text{gauss})/K(\varphi_0) = J_0$ . Thermodynamics fixes the unit of cost (remark 4.16) and motivates this framework, without entering into a proof.

**Proposition 4.23** (admissibility of  $\mathcal{F}$ ).  $\forall c > 0: cJ \in \mathcal{C}$ : positivity and  $\tau$ -l.s.c. by assumption 9; global domination  $cJ \geq \varepsilon J$  with  $\varepsilon = c$ . Hence  $\mathcal{F} \subset \mathcal{C}$  and all the machinery (Section 4.2) and the wall (theorem 4.10) apply.

## 5 Consequences: a well-posed inference

The wall (theorem 4.10) pushes the boundary of certainties  $\partial_\infty$  to infinite  $d_e$ -distance: every curve approaching it has a cost length  $\ell_e$  that diverges. Three consequences follow: a minimizer exists without collapsing, the cost bounds the entropy variation, and reaching a precision has a geometric cost floor.

**Corollary 5.1** (existence and attainability of the minimizer). *By theorem 4.10, the cost length  $\ell_e$  diverges on approach to  $\partial_\infty$ : it is coercive on the candidate class, and the cost metric is geodesically complete there — the divergence of  $\ell_e$  at the boundary (theorem 4.10) together with the compactness of the sublevels  $\{m_2 \leq M\}$  (assumption 6) provides it, assumption 8 (Hopf–Rinow) being invoked only on these locally compact sublevels, not on all of  $\mathcal{P}_2$ : no geodesic reaches the boundary at finite length. A minimizing sequence then stays in a sublevel  $\{m_2 \leq M\}$ ,  $\tau$ -sequentially compact; Arzelà–Ascoli extracts a limit, which the lower semicontinuity of  $\ell_e$  selects as a minimizer in the interior of the domain (theorem 4.5). The minimizer exists and does not collapse onto certainty.*

**Corollary 5.2** (Lipschitz control of the cost in entropy). *By theorem 4.10, the boundary is at infinite  $d_e$ -distance: no curve of finite cost length reaches it, and every pair of beliefs in  $\mathcal{R}$  stays at finite  $d_e$ -distance. The Lipschitz property of the entropy ( $|\Delta H| \leq \frac{1}{\sqrt{2\varepsilon}} d_e$ , theorem 4.10(i)) bounds the entropy variation along any curve by its cost length: each unit of length can lower the entropy by at most  $\frac{1}{\sqrt{2\varepsilon}}$ . This control bears only on the entropic term: the data-attachment and interaction contributions are not bounded by the wall.*

**Corollary 5.3** (geometric cost floor). *By theorem 4.10(i), every curve joining two beliefs whose entropies differ by  $\Delta H$  has a  $d_e$ -length at least  $\sqrt{2\varepsilon} |\Delta H|$ . Reaching a given precision (target variance) thus has a geometric cost floor. At the wall,  $d_e \geq \sqrt{2c} |\Delta H|$  diverges with  $|\Delta H|$ . (The energetic reading of this floor — its proportionality to the Landauer energy at the boundary — is deferred to remark 4.16, tier-P, outside the proof.) This floor is purely geometric, in nats; its value depends on the cost unit, and its divergence at certainty is invariant of it (theorem 4.21).*

**Existence without uniqueness.** Completeness and coercivity give the existence of a minimizer, not its uniqueness. The latter would require the geodesic convexity of the objective in the cost metric, not established in general (Section 6). The framework establishes well-posedness *for existence and stability*; uniqueness remains open — the non-convexity of possible interaction terms, and the multiplicity of minima it may induce, are left open.

## 6 Discussion: scope and limits

Three results are established: a wall, an eikonal equivalence, a curvature bounded below by Stam. It remains to situate them — with respect to what stays conjectural and to what we do not claim (below); the positioning relative to neighboring geometries is treated together with the related work (Section 7).

### What remains conjectural

Three statements bound the scope of the theorems; we fix their exact status, without broadening it.

*Remark 6.1* (C1 — beyond the conformal class). theorem 4.10(i) (sufficiency) holds for every  $U \in \mathcal{C}$ ; theorem 4.10(ii) (necessity) is proved on the power family  $\{cJ^\alpha\}$ . **Conjecture C1.** (a) Necessity extends to every  $U \in \mathcal{U}$  (any  $U$  not dominating  $J$  near  $\partial_\infty$  leaves the boundary at finite distance). (b) Beyond conformal deformations (definition 2.6), the characterization remains open. *We never claim uniqueness outside the conformal class.*

*Remark 6.2* (C2 — curvature beyond the Gaussian: resolved on location-scale). **Resolved (location-scale).** Beyond the Gaussian leaf, theorem 4.18 establishes that on *every* 1D location-scale family the curvature equals  $-1/(2cJ_0) < 0$ : *negativity is robust*, and the extremal value is attained by the Gaussian (Stam,  $J_0 = 1$ ). **Conjecture C2 (what remains open).** (a) Negativity near  $\partial_\infty$  persists on 2D families that are *not* location-scale (curved  $\mathcal{W}_2$  base). (b) In dimension  $\geq 2$  / on all of  $\mathcal{P}_2$ , the *ambient sectional* curvature of  $\tilde{g}_{e,2J}$  is negative near  $\partial_\infty$  — a heavier object: Otto computation + *dimension-dependent* conformal transformation. *Tooling note:*  $CD(K, N)$  (Lott and Villani, 2009; Sturm, 2006) bounds the Ricci of the *base* ( $\mathcal{P}_2, \mathcal{W}_2$ ) via convexity of the entropy; this is *not* the curvature of the cost metric  $\tilde{g}$  (a distinct conformal object) —  $CD(K, N)$  therefore applies only to part (b), not to theorem 4.18. *We do not conflate the intrinsic family curvature (theorem 4.18) and the ambient sectional one.*

*Remark 6.3* (C3 — global existence on  $\mathcal{P}_2$ ). theorem 4.5 gives existence for the action; the existence of the minimizer of an estimate on all of  $\mathcal{P}_2(\mathbb{R}^n)$  (mass escape) requires concentration-compactness (Lions, 1984), beyond the reach of a properly proved statement as it stands. *Status:* scoped; the interior minimizer guaranteed by the wall (theorem 4.10) suffices for fixed endpoints / coercive fidelity.

### Acknowledged limits

We claim no more than what is established: the curvature *reads* the Fisher information, it does not ground it; the value  $-\frac{1}{4}$  is the image of a unit (remark 4.20); “well-posed” covers only existence and stability; and we characterize only a *family*, within the *conformal* class. The four open points — general necessity, universality of the curvature, global existence on  $\mathcal{P}_2$ , uniqueness of the minimizer — are the conjectures scoped above (remarks 6.1 to 6.3).

## 7 Related work

Each building block is cited where it is used; we gather here the boundary between what is acquired and what is contributed, complementing the three positioning remarks (remarks 7.1 to 7.3). The

contribution amounts to two moves: the eikonal characterization  $U = cJ$  (theorem 4.14) and the assembly into a cost metric with a wall; the rest is borrowed.

**Geometry of inference.** Information geometry (Chentsov, 1972; Amari, 2016) equips beliefs with the Fisher–Rao metric, the only one invariant under statistical morphisms. Otto’s calculus (Otto, 2001) makes the complementary move: it endows  $\mathcal{P}_2$  with a Riemannian structure of distance  $\mathcal{W}_2$ , where the heat flow is the gradient flow of entropy (Jordan et al., 1998; Ambrosio et al., 2008) and where  $\|\nabla_{\mathcal{W}_2} H\|^2 = J$  (Villani, 2009) — the identity that makes  $J$  the *slope* of  $-H$  (assumption 4), not a tensor. The HWI inequality (Otto and Villani, 2000) already relates our objects in one line. Our framework characterizes *another* class, the conformal deformations of transport, whose curvature signature  $-\frac{1}{4}$  separates it from Fisher–Rao (remark 7.1).

**Thermodynamics of information.** Erasing one nat dissipates at least  $k_B T$  (Landauer, 1961; Bennett, 1982); the thermodynamics of information turned this into an exact bookkeeping (Parrondo et al., 2015) — extractable work bounded by the information acquired (Sagawa and Ueda, 2010), and an *energetic* reading of geodesics where the path of minimal dissipation is optimal transport (Aurell et al., 2011; Dechant and Sakurai, 2019). This is the currency, nat for nat, and the legitimacy of treating an inference device as a thermodynamic system; the energetic optimality of geodesics is their theorem, not ours. Our cost is *derived* from the right price.

**Finite-time dissipation.** Benamou–Brenier (Benamou and Brenier, 2000) gives  $\mathcal{W}_2^2$  a kinetic reading; stochastic thermodynamics turns it into a price — the dissipation of a diffusion driven from  $p_0$  to  $p_1$  in time  $T$  is bounded below by  $\mathcal{W}_2^2(p_0, p_1)/T$  (Dechant and Sakurai, 2019), and the thermodynamic length (Sivak and Crooks, 2012) is its linear regime; a second family bounds the *rates* by the Fisher *of the path* (Ito, 2018). These works *measure* the dissipation on a given geometry; they do not re-metrize the space of beliefs nor place a wall in it, and their Fisher is that of the path, not of the state  $J(p)$  (remark 7.2).

**Fisher cost in learning.** This is a direct link. Elastic weight consolidation (Kirkpatrick et al., 2017) fights forgetting by penalizing the displacement of a belief by a *squared Fisher distance* toward a reference state, derived from a Laplace approximation of the posterior (Huszár, 2018): on the Gaussian leaf, the same object as our maintenance cost, but *static* — a frozen anchor — where the maintenance cost is dynamic. Still et al. (2012) price *non-predictive* information under driving and Kolchinsky and Wolpert (2018) ground semantic information thermodynamically; here even useful information has a permanent maintenance cost, because in our setting the world diffuses. The natural gradient (Amari, 2016) preconditions by the Fisher, a relative of well-posedness (corollary 5.1).

**Mixed Wasserstein–Fisher geometries, and positioning.** von Renesse (2012) reads the Schrödinger equation as a Newton law on  $\mathcal{P}_2$  with potential the Fisher, and Conforti and Pavon (2018) classify transport, Schrödinger bridge and Madelung fluid as extremals of actions that differ only by the *sign* of a Fisher term; to be distinguished from the Wasserstein–Fisher–Rao metric (Ambrosio and Santambrogio, 2007; Chizat et al., 2018), which interpolates *additively* by varying the mass, the opposite of our conformal deformation at mass 1. Concurrent works (2024–2026) treat neighboring facets without covering this assembly: Okanohara (2026) bounds the dissipation of learning by an ensemble Wasserstein distance (transport of models, a relative of that of Dechant and Sakurai (2019), where the maintenance cost bears on the *state* of Fisher  $J(p)$ ), Melo et al. (2025) study the entropy production tied to Fisher information in stochastic thermodynamics, of which Ito and Sagawa (2016) gives the Bayesian-network setting, and Hyland and Albarracín (2025) price the cost of *changing* belief, where the wall (theorem 4.10) prices that of *holding* it.

To our knowledge, the eikonal characterization  $U = cJ$  and the metric inaccessibility of certainty as a wall on the conformal class have not been formulated as such.

## Positioning

The characterization is read first by contrast, opposing three currencies — Fisher–Rao, bare transport and honest cost — on the same location-scale leaf; the three remarks that follow unfold the argument, each against an established program.<sup>2</sup>

*Remark 7.1* (vs information geometry (Amari–Čencov)). Čencov characterizes Fisher–Rao as *the unique* metric invariant under statistical morphisms (Chentsov, 1972; Amari, 2016). Our characterization is of analogous nature but in *another class*: conformal deformations of *transport* (Wasserstein), where  $J$  is the *slope* of  $-H$  (assumption 4), not the metric tensor. Distinctive signature: curvature  $-\frac{1}{4}$  (representative) *vs*  $-\frac{1}{2}$  of Fisher–Rao (Atkinson and Mitchell, 1981; Costa et al., 2015). On location-scale families, Fisher–Rao reads the *scale* information  $b_0^2$  (Gaussian  $-\frac{1}{2}$ , with no link to Stam) (Nielsen, 2021) and *bare* transport has curvature  $\geq 0$  (Takatsu, 2011), where the *honest* cost makes the *position* information  $J_0$  carry the curvature,  $J_0$  being bounded below by Stam (theorem 4.18 and remark 4.19) — the negativity arises from the conformal deformation, not from bare transport. *As with Čencov, the uniqueness (here of family) is relative to the class; we make it explicit.*

*Remark 7.2* (vs geometric thermodynamics (Ito)). The Fisher $\leftrightarrow$ Wasserstein bridge in stochastic thermodynamics is established (Sivak and Crooks, 2012; Aurell et al., 2011; Ito, 2023): the entropy production there is bounded by  $\mathcal{W}_2$  (*speed limit*),  $J$  appearing as the excess entropy rate. These works *measure* the dissipation on a given geometry; they do not re-metrize the space of beliefs nor place a wall in it. We *characterize* a class of conformal *cost* metrics on the beliefs, endowed with a wall and a curvature.

*Remark 7.3* (vs Bayesian mechanics (Friston)). The slogan “dissipative agent  $\rightarrow$  geometry of belief” is carried by the free-energy principle (Friston, 2019; Sakthivadivel, 2022), on a Fisher manifold. These approaches introduce neither a wall, nor negative curvature driven by *position* information, nor an eikonal characterization  $U = cJ$  of a cost class on Wasserstein. The *delta* of this work lies in this *assembly* — the eikonal and the wall on transport — more than in the novelty of each ingredient taken in isolation.

## 8 Perspectives

The preceding characterizations concern a *background* cost geometry: fixed by honesty (theorem 4.14) and determined up to a unit (theorem 4.21), the metric  $\tilde{g}_{e,U} = 2(e + U) g_{\mathcal{W}_2}$  is a scene given in advance, where a belief moves like a test particle. A *self-consistent* geometry, sourced by the configuration of beliefs it carries rather than prescribed, could extend it. The displacement would then bear on the scene itself, beyond the cost unit fixed by gauge invariance (theorem 4.21); we borrow from general relativity, as an analogy, its lesson of background independence (Einstein, 1916). One ingredient already lends itself to this: in the kinetic action  $\frac{1}{2}|\dot{\gamma}|_{\tilde{g}}^2 = (e + cJ)|\dot{\gamma}|_{\mathcal{W}_2}^2$ , the factor  $(e + cJ)$  would play the role of an *inertial mass* — the resistance to changing one’s mind, large near the wall (theorem 4.10). It remains to define the *active* mass of a belief — by which it could, in return, contribute to the geometry seen by other beliefs — and its coupling. The two statements below, in the conditional, mark out the way<sup>3</sup>; following the convention of the paper, they *motivate* and enter into no proof.

<sup>2</sup>The location-scale leaves are also the noise models of M-estimators: a standardized form  $\varphi_0$  is the loss  $\rho = -\log \varphi_0$ , and the curvature  $K = -1/(2cJ_0)$  (theorem 4.18) orders these estimators, the Gaussian (least squares) being the Stam extreme — a bridge to robust estimation (Huber, 1981), not exploited here.

<sup>3</sup>C4 and C5 follow the numbering style of conjectures C1–C3 (remarks 6.1 to 6.3); unlike those, which bound the scope of the established theorems, C4–C5 open directions outside the proof structure.

*Remark 8.1* (C4 — active mass and inertial mass). In a self-consistent completion of the cost geometry — where the relief  $U$  would be sourced by the belief configuration —, the active mass of a belief would coincide with its inertial mass  $\propto (e + cJ)$ , already present as the factor of  $|\dot{\gamma}|^2$  in  $\tilde{g}$ .

*Remark 8.2* (C5 — toward an epistemic hysteresis). A nonconvex attractive interaction term between correlated beliefs would induce a multiplicity of minima; under a quasi-static dynamics, the belief would exhibit a *hysteresis*, whose loop width would grow with the inertial mass ( $\propto e + cJ$ ) and the coupling. The two-route first-order transition signaled in theorem 4.7 would be its minimal instance.

The development of these directions — self-consistent geometry, interaction dynamics, hysteresis — belongs to separate work.

## 9 Conclusion

In the setting we consider, a finite agent infers the state of the world through a noisy sensor: we model its output as a belief, and two forms of finiteness forbid it certainty. On the conformal class, the wall — certainty pushed off to infinite distance — is *characterized* by the dominance of Fisher information (sufficient unconditionally; necessary at the boundary, general case conjectured; theorem 4.10); there the eikonal selects the Fisher family  $U = cJ$  (theorem 4.14), and the curvature  $K = -1/(2cJ_0)$ , *over any location-scale family*, is extremal at the Gaussian, by Stam rigidity (theorems 4.17 and 4.18). These three characterizations are the invariants of one and the same change of cost unit (theorem 4.21): thermodynamics fixes the unit, and the proven conclusions hold up to that unit. We thus obtain, within the conformal class, a *family* over which inference is well-posed, and reaching a precision has a floor that diverges at certainty. Its extension — a geometry sourced by the beliefs it carries — is sketched in Section 8.

## A Proofs of the structural results

We prove here the class-machinery statements announced in Section 4.2 — the AM–GM inequality (lemma 4.3), the Maupertuis–Jacobi correspondence (theorem 4.4), the existence of geodesics (theorem 4.5), the energy law (proposition 4.6) and the equivalence optimal curve  $\Leftrightarrow$  geodesic (theorem 4.7) —, valid for every candidate  $U \in \mathcal{U}$ ; the regime clauses that bound them (duality gap  $e > 0$ , du Bois-Reymond lemma, compactness of the sublevels by Arzelà–Ascoli) enter explicitly in each proof.

*Proof of lemma 4.3.*  $\frac{1}{2}a^2 + b - \sqrt{2b}a = \frac{1}{2}(a - \sqrt{2b})^2 \geq 0$ , zero iff  $a = \sqrt{2b}$ . □

*Proof of theorem 4.4.* (i) lemma 4.3 with  $a = |\dot{\gamma}|$ ,  $b = e + U$ , integrated. (ii) reparametrize at constant  $\mathcal{W}_2$  speed  $L$  (Ambrosio et al., 2008, lem. 1.1.4), set  $t(s) = \int_0^s L/\sqrt{2(e + U(\hat{\gamma}_r))}dr$  (integrand  $\leq L/\sqrt{2e}$ ,  $> 0$  a.e.); the equality case of lemma 4.3 concludes. (iii) equality in (i) = pointwise equality case. □

*Proof of theorem 4.5.* Minimizing sequence  $\mathcal{A}_T(\gamma^k) \leq C$ ; lemma 4.1(a) gives  $K(\gamma^k) \leq 2C$ , hence Hölder equicontinuity for the Dudley metric  $\beta_0 \leq \mathcal{W}_1 \leq \mathcal{W}_2$ , and  $m_2(\gamma_t^k) \leq M$ : the curves stay in the  $\tau$ -compact  $\{m_2 \leq M\}$  (assumption 6). Arzelà–Ascoli + l.s.c. ( $K$  by lemma 4.1(c);  $\int U$  by l.s.c. of  $U$  and Fatou) give  $\mathcal{A}_T(\gamma) \leq \Phi(T)$ . □

*Proof of proposition 4.6.* Inner variations: for  $u : [0, T] \rightarrow (0, \infty)$ ,  $u, 1/u \in L^\infty$ ,  $\int u = T$ ,  $\gamma_t^u := \gamma_{\theta_u(t)}^*$  is admissible and  $\mathcal{A}_T(\gamma^u) = \int [\frac{1}{2}|\dot{\gamma}^*|^2/u + Uu] =: F(u)$ .  $F$  convex,  $u \equiv 1$  optimal.  $V(\sigma) := \inf\{F : \int u = \sigma\}$  convex;  $\text{dom } V = (0, \infty)$  (for every  $\sigma > 0$ ,  $u_\sigma := \sigma/T$  gives  $\int u_\sigma = \sigma$  and  $F(u_\sigma) = \frac{T}{\sigma} \int \frac{1}{2}|\dot{\gamma}^*|^2 + \frac{\sigma}{T} \int U(\gamma^*) < \infty$ ), so  $T$  is *interior* there and  $\partial V(T) \neq \emptyset$ ; taking  $-e \in \partial V(T)$ , the first variation  $\int [-\frac{1}{2}|\dot{\gamma}^*|^2 + (U + e)]h = 0$  ( $\forall h \in L^\infty$ ) gives (du Bois-Reymond) the stationarity, and the convexity of  $u \mapsto \frac{1}{2}|\dot{\gamma}^*|^2/u + (U + e)u$  ( $\partial_u^2 \geq 0$ ) makes it the pointwise minimizer. □

**Table 2: Dependency table** (authoritative). Provenances: **(P)** physical/interpretive (motivates, never a proof); **(E)** borrowed (standard theorem cited); **(D)** proved here, in pure metric geometry.

| Statement                                      | Tier                | Direct logical dependencies                                                                           |
|------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------|
| P: physical (Landauer/Bekenstein)              | <b>(P)</b>          | — ( <i>motivates only, dashed; operational constraints posed as setting, remark 2.3</i> )             |
| A1, A2, A3; B1–B7                              | setting/ <b>(E)</b> | —                                                                                                     |
| lemma 4.3, lemma 4.1                           | <b>(D)</b>          | —; B3                                                                                                 |
| lemma 4.2                                      | <b>(E)</b>          | B1, B2                                                                                                |
| Maupertuis (theorem 4.4)                       | <b>(D)</b>          | lemma 4.3                                                                                             |
| existence (theorem 4.5)                        | <b>(D)</b>          | lemma 4.1, B3                                                                                         |
| energy law (proposition 4.6)                   | <b>(D)</b>          | theorem 4.5                                                                                           |
| geodesic (theorem 4.7)                         | <b>(D)</b>          | theorem 4.4, proposition 4.6                                                                          |
| <b>Char. I</b> wall (theorem 4.10)             | <b>(D)</b>          | definition 4.8, lemma 4.2, B1, B4, theorem 4.5                                                        |
| <b>Char. II</b> eikonal (theorem 4.14)         | <b>(D)</b>          | B1, definition 4.12                                                                                   |
| admissibility $\mathcal{F}$ (proposition 4.23) | <b>(D)</b>          | B6                                                                                                    |
| hyperbolicity (theorem 4.17)                   | <b>(D)</b>          | definition 4.13, B4                                                                                   |
| Stam extremum (theorem 4.18)                   | <b>(D)</b>          | B7, B6 (Stam)                                                                                         |
| representative $2J$ (remark 4.20)              | <b>(D)</b>          | theorem 4.14, theorem 4.17, theorem 4.18                                                              |
| setting defs.                                  | setting             | definitions 2.6, 4.8, 4.9, 4.12 and 4.13 — ( <i>e and the “2” enter as a unit; cf. theorem 4.21</i> ) |
| gauge invariance (theorem 4.21)                | <b>(D)</b>          | definition 2.6, theorem 4.18                                                                          |
| finiteness cor. (corollary 4.11)               | <b>(D)/(P)</b>      | theorem 4.10; <i>motivates</i> : P (dashed)                                                           |
| honesty of the price (remark 4.16)             | <b>(D)/(P)</b>      | theorem 4.10, theorem 4.14, theorem 4.17; <i>motivates</i> : P (dashed)                               |
| base (definition 2.1, definition 2.2)          | setting             | — ( <i>object: <math>H, J, \partial_\infty</math></i> )                                               |
| existence (corollary 5.1)                      | <b>(D)</b>          | theorem 4.10, theorem 4.5, assumption 8                                                               |
| cost–entropy (corollary 5.2)                   | <b>(D)</b>          | theorem 4.10(i)                                                                                       |
| cost floor (corollary 5.3)                     | <b>(D)/(P)</b>      | theorem 4.10(i), theorem 4.17; <i>motivates</i> : P (dashed)                                          |

*Proof of theorem 4.7.* The energy law gives the pointwise equality of lemma 4.3, hence  $\mathcal{A}_T(\gamma^*) + eT = \ell_e(\gamma^*)$  and the optimality of the trace for  $\ell_e$  at parameter  $T_e = T$ . The strict concavity of  $x \mapsto \ell_x(\gamma^*) - xT$  ( $G'' < 0$ ) and exact duality force  $\ell_e(\gamma^*) = \Psi(e)$  (global geodesic). Converse: theorem 4.4(ii).  $\square$

## B Dependency table and logical independence of the proofs

table 2 records, statement by statement, the authoritative direct logical dependencies; it is the only place where the provenances P (physical/interpretive), E (borrowed) and D (proved) are labeled, following the convention recalled in the table’s note. The acyclicity proposition below establishes that no proved statement depends on a physical statement.

**Proposition B.1** (acyclicity of dependencies: no proved statement depends on a physical statement **(D)**). *Let  $\rightarrow$  denote direct logical dependence (col. 3 of table 2, which is authoritative; set-theoretic reading:  $P$  outside the universe). Then: (i) the graph on  $\{P\} \cup \{\text{setting}\} \cup \{\text{borrowed results (E)}\} \cup \{\text{statements (D)}\}$  is acyclic (stratification by rank: each solid edge goes from a lower rank to a higher rank); (ii)  $\forall D \text{ (D)} : P \notin \text{cl}_{\rightarrow}(D)$ :  $P$  has no outgoing solid edge (the “motivates” col. = dashed edges only), so every  $D$  is proved from setting + borrowed results alone. Each edge of table 2 has been traced back in the body of the corresponding proof (finite verification).*

*Proof.* By inspection of table 2: no “logical dependencies” column contains P; the rank ( $\text{rank}(D) = 1 + \max\{\text{rank}(x) : x \rightarrow D\}$ ) is well-defined and finite, whence acyclicity; induction on the rank for  $P \notin \text{cl}_{\rightarrow}(D)$ .  $\square$

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